Differentially Private Network Analysis

Main Paper: Inference Using Noisy Degrees, Karwa and Slavković [2016]

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Contents

0.1 Main objective

Concept of data privacy has become crucially important for statistics and data analytics research. Recently, models and learning algorithms have become extremely competitive with increasing amount of data. However, there is also a growing concern of protection of sensitive information. Actually, this has become a main discussion point for all data analytics professionals and researchers since the Cambridge Analytica, and Facebook scandal in 2018, especially after government involvements. Recently, both US and European Union issued executive orders to take precautions against unsafe use of statistical and computational models. Additionally, network analysis models have been in use for a long time to model complex relationships. Given these facts, this paper proposed a method to solve a maximum likelihood estimation problem for a network model under privacy constraints. For this purpose, authors used a well-known network model, β-model (also called as exponential random graph models), and a popular privacy mechanism, Differential Privacy. To understand the problem and the method, one needs to understand these two techniques.

Firstly, β-network models assign a probability of having a a connection between edge using a logistic function with β parameter. This leads to the fact that one can estimate these parameters just using a degree sequence of the graph. Note that degree sequence is a sequence which has size of total number of nodes, and each value is determined by total number of nodes connected to that certain node.

Secondly, differential privacy usually acts as a noise injection mechanism. In other words, to make a randomized algorithm differentially private, usually output of the algorithm is corrupted with some carefully chosen noise to satisfy the privacy constraint. Therefore, a natural extension might be adding a noise on degree sequence of the graph and using that noisy sequence for estimating β parameters. However, this idea is infeasible, and it creates problems for estimation. One issue is that noise mechanisms can usually add negative noise on the output of the algorithm. This means that degree sequence might become negative after noise injection, which must be impossible in theory. Additionally, MLE of β exists under certain conditions on degree sequences. However, after noise injection, these conditions might not be satisfied anymore. This paper developed a methods to find easier and efficient way to check existence of MLE, and utilizing noisy degree sequence without loosing MLE and consistency.

0.2 Main Tools

One can summarize the contributions of the paper as follows.

1. **Necessary and sufficient conditions of MLE:** Authors proposed computationally more efficient conditions for checking existence of MLE in β network models. Current technique from Rinaldo et al. [2013] is more general but less tractable. These results apply beyond privacy context.

2. **Directly using noisy statistic is problematic:** In a β model directly utilizing noisy statistic without considering noise mechanism may result in nonexistence of MLE. Also, L1 and L2 minimization is not enough for statistical utility guarantee. One should consider noise mechanism explicitly.

3. **Private MLE estimate of $\hat{d}$:** One can obtain MLE estimates of $\hat{d}$ (degree sequences) from its noisy counterpart z. This is a non-standard MLE estimation procedure with discrete parameter space and increasing dimensionality with increasing sample size.

4. **DP estimation of β using $\hat{d}$:** Then we can come up with consistent and asymptotically normal estimator $\hat{\beta}_\epsilon$ of β.

Current methods for checking existence of MLE are computationally expensive [Chatterjee et al., 2011]. Therefore, authors proposed a more efficient way of checking this by considering the convex hull and interior of convex hull of the all graphical degree sequences. Using the properties of convex hulls, they came up with simple and easier-to-implement conditions.

In addition to that, paper proposed a method to utilize noisy sequence for MLE estimation. This method mainly relies on two algorithms. First algorithm is corrupting the original sequence using Discrete Laplace Mechanism. This version of Laplace mechanism add integer (either positive or negative) noise on degree sequences and make it differentially private. By the properties, once a differential privacy mechanism is utilized, it is safe to use the output without being concerned about possible privacy leakage. Therefore, main idea here is that making the degree sequence noisy so that it is impossible to recover original graph exactly, and then use this noisy sequence for MLE estimation. Second algorithm is developed to mitigate the problems of using noisy sequences. Roughly, this algorithm recovers a graph using noisy sequence by finding a graph which has a degree sequence closest to the given noisy sequence in the sense of $L1$ norm. In a way, this algorithm finds the MLE of degree sequence (which satisfies the conditions for MLE existence). The idea behind this algorithm stems from the Havel-Hakimi decomposition procedure [Erdős et al., 2009]. Here, note that the resulting graph may be disconnected, it means that MLE doesn't exist. One solution is provided in the results section.

Finally, authors showed that MLE estimation of β parameter using MLE of degree sequence found by graph recovering algorithm is consistent and asymptotically normally distributed.

0.3 Main results & Proofs

0.3.1 Setup

<u>**Notation:**</u>

G_n : Simple undirected graphs with n nodes and m edges
V : Vertex set
E : Edge set
$A : \{i \in n, (i,i) \notin E, |(i,j) : (i,j) \in E| = 1\}$(simple graph)
$\delta(G, G')$: Number of edges on which the graphs differ
d_i : Number of nodes connected to node i

Definition 1 (Degree sequence and partition) *Sequence of degrees is called degree sequence and denoted as $d = \{d_1, d_2, \ldots, d_n\}$. Furthermore, degree sequence ordered in nonincreasing order is called degree partition and denoted by $\bar{d} = \{d_{(1)}, d_{(1)}, \ldots, d_{(n)}\}$.*

Given the degree sequence, we can define

$\mathcal{G}(d)$: *Set of simple graphs on n vertices with degree sequence d*
DS_n : *Set of all graphical degree sequences of size n*
DP_n : *Set of all graphical degree partitions of size n*
$\mathcal{U}(d)$: *Uniform distribution over the set of all graphs with degree sequence d*

where "graphical degree sequences" refers to the sequences that can be realized by a simple graph.

Definition 2 (ϵ-Differential Privacy Dwork [2008], Dwork et al. [2014]) *A randomized algorithm M satisfies ϵ-differential privacy if for all datasets D_1, D_2, which satisfy $\|D_1 - D_2\|_H = 1$, and $S \subseteq Range(M)$*

$$Pr[M(D_1) \in S] \leq e^{\epsilon} \times Pr[M(D_2) \in S]$$

where $\|D_1 - D_2\|_H = \sum_{i=1}^{N} 1_{D \neq D'}$

Inference with β model: Consider $\beta = \{\beta_1, \beta_2, \ldots, \beta_n\} \in \mathbb{R}^n$ as a fixed point. For a random graph on n vertices, probability of having an edge between nodes i and j (for $i < j$) is independently

$$p_{ij} = \frac{e^{\beta_i + \beta_j}}{1 + e^{\beta_i + \beta_j}}$$

This characterization is a special case of discrete exponential family with degree sequence d as a sufficient statistic, i.e.

$$P(G = g) \propto \exp \sum_{i=1}^{n} d_i \beta_i$$

0.3.2 Results

First result is related to the first part of the estimation procedure, which makes the degree sequences noisy. This result shows that discrete Laplace distribution can be used for differential privacy purposes.

Result 1 (Edge Differential Privacy) *Let $Z_1, Z_2, \ldots, Z_k$ be i.i.d discrete Laplace random variables with p.m.f defined as follows:*

$$P(Z = z) = \frac{1-\alpha}{1+\alpha}\alpha^{|z|}, \quad z \in \mathbb{Z}, \alpha \in (0,1)$$

Then the algorithm which outputs $f(G) + (Z_1, Z_2, \ldots, Z_k)$ for graph G is ϵ-edge differentially private where $\epsilon = [-\Delta(f)\log\alpha]$ and $\Delta(f)$ is global sensitivity of f. Edge differential privacy is defined as following. If an algorithm satisfies this constraint, similar to the above definition, it is edge differentially private.

$$\sup_{\mathcal{G},\mathcal{G}',\delta(\mathcal{G},\mathcal{G}')=1} \log\frac{\mathcal{Q}(S|\mathcal{G})}{\mathcal{Q}(S|\mathcal{G}')} \leq \epsilon$$

where S is the set of all possible outputs, and $\epsilon > 0$.

Proof 1 (Edge Differential Privacy) *Let g and g' be two graphs that differ by an edge. For any $s \in \mathbb{Z}^k$ and $\alpha \in (0,1)$, consider*

$$\begin{aligned}
\mathbb{P}[f(\mathcal{G}) + Z = s|\mathcal{G} = g] &= \mathbb{P}[Z = s - f(\mathcal{G})|\mathcal{G} = g] \\
&= \prod_{i=1}^{k}\mathbb{P}[Z_i = s_i - f_i(g)|\mathcal{G} = g] \\
&= c(\alpha)\alpha^{\sum_{i=1}^{k}|s_i - f_i(g)|}
\end{aligned}$$

Then

$$\begin{aligned}
\frac{\mathbb{P}[f(\mathcal{G}) + Z = s|\mathcal{G} = g]}{\mathbb{P}[f(\mathcal{G}) + Z = s|\mathcal{G} = g']} &= \alpha^{\sum_{i=1}^{k}|s_i - f_i(g)| - |s_i - f_i(g')|} \\
&\leq \alpha^{-\sum_{i=1}^{k}|f_i(g) - f_i(g')|} \\
&\leq \alpha^{-\Delta(f)} \\
&= \exp(\epsilon)
\end{aligned}$$

where second inequality follows from the triangular inequality, $|s_i - f_i(g)| - |s_i - f_i(g')| \geq -|f_i(g) - f_i(g')|$. This means that $\sum_{i=1}^{k}|s_i - f_i(g)| - |s_i - f_i(g')| \geq -\sum_{i=1}^{k}|f_i(g) - f_i(g')|$, and since α^x is decreasing with respect to x when $\alpha \in (0,1)$, we have $\alpha^{\sum_{i=1}^{k}|s_i - f_i(g)| - |s_i - f_i(g')|} \leq \alpha^{-\sum_{i=1}^{k}|f_i(g) - f_i(g')|}$. Third inequality follows from the fact that

$$\Delta(f) = \sup_{g,g':\|g-g'\|_H=1}\|f(g) - f(g')\|_1 = \sup_{S_i}\sum_{\{S_i:\|g_i - g_i'\|_H=1\}}|f(g_i) - f(g_i')|.$$

this means that $\sum_{i=1}^{k}|f_i(g) - f_i(g')| \leq \Delta(f)$, i.e. $-\sum_{i=1}^{k}|f_i(g) - f_i(g')| \geq -\Delta(f)$, so $\alpha^{\sum_{i=1}^{k}|s_i - f_i(g)| - |s_i - f_i(g')|} \leq \alpha^{-\Delta(f)}$.

Other main technique is for recovering a graph from the noisy sequence. However, it is very detailed and not closely related to the topics discussed in the class. Therefore to be concise, it is omitted. However, one can check details from Karwa and Slavković [2016], Algorithm 2. Similarly, proof is also off the topic and related to convex hull operations. However, to give an idea to the reader, a rough sketch of the proof is provided here.

Result 2 (Validation of Algorithm 2) *MLE estimation of degree sequence d, which is $\hat{d}$, calculated using the graph outputted by Algorithm 2, from Karwa and Slavković [2016], is in DS_n and $\hat{d} \in ri(Conv)(DS_n)$.*

Proof 2 (Sketch of the proof for validation of Algorithm 2) *Proof requires 3 steps starting with defining the MLE problem as L_1 projection on the set DS_n, which is $\hat{d} = arg\min_{d \in DS_n} ||d - z||_1$.*

1. *First construct an iterative way to check if resulting sequence is in DS_n, and then using this procedure show that every degree sequence can be described as sum of k-star degree sequences. Note that k-star degree sequence describes a graph centered at some node with exactly k-edges connected to that node.*

2. *Then using Havel-Hakimi decomposition (which is simply set of k-star degree sequences that forms the sum in the previous part), show that to find an optimal degree sequence $\hat{d}$, it is enough to consider sequences pointwise bounded by z.*

3. *Finally show that one can reduce L_1 distance of any degree sequence by replacing k-star sequences of its Havel-Hakimi decomposition with appropriately chosen sequences.*

Algorithm 2 simply applies this steps sequentially, and therefore it produces a graph that realizes the degree sequence of L_1 projection. Note that algorithmic complexity for this algorithm is $O(n \log n + m)$ where $n \log n$ comes from the sorting step and m comes from adding edges.

As it was mentioned before, the graph recovering algorithm (which was proved roughly above) usually produces a disconnected graph, which makes it impossible to find the MLE estimation β. In this case one can convert the disconnected graph to connected graph with the following operation. This new sequence is also MLE of d. Note that disconnectedness means that certain nodes have degree 0.

Result 3 (Connecting graph) *Let d^* be the output of algorithm, which is a degree sequence of an disconnected graph. Let $S1 = \{j : d_j^* = 0\}$, i.e indexes of disconnected nodes. Also, take $S2 = \{j : d_j < z_j, d_j > 0\}$, i.e. indexes where we add undesirably large noise. Then define*

$$d_i = \begin{cases} 1 & i = j \\ d_i^* + 1 & i = k \\ d_i^* & otherwise \end{cases}$$

for $j \in S1$, and $k \in S2$. In this case, we have $\|d - z\| = \|d^* - z\|$.

Proof 3 (Connecting graph) *First of all, Algorithm 2 constraints the number of edges connected to a node in order to prevent failures. This is because z can get grow larger than graph because the noise is random. So, in that sense, we have $d_i^* \leq z_i$, $\forall i$. Before proving the inequality, we need following claim.*

Claim: *$z_j \leq 0$ if and only if $d_j^* = 0$.*
Proof: *If $z_j \leq 0$ then $d_j^* = 0$ is trivial by the design of algorithm. If $d_j^* = 0$, this means that $\|d^* - z\|$ is minimized for j when $d_j^* = 0$. Since we add discrete noise, z_j is integer. Therefore, $d_j^* = 0$ minimizes the norm when $z_j \leq 0$.*

Therefore, for $j \in S1$, and $k \in S2$, we have

$$
\begin{aligned}
\|d - z\| &= \sum_i |d_i - z_i| = \sum_{i \setminus \{j,k\}} |d_i - z_i| + |d_j - z_j| + |d_k - z_k| \\
&= \sum_{i \setminus \{j,k\}} |d_i^* - z_i| + |1 - z_j| + |1 + d_k^* - z_k| \\
&= \sum_{i \setminus \{j,k\}} |d_i^* - z_i| + 1 - z_j + |1 + d_k^* - z_k| \quad (\textit{by the definition of S1}) \\
&= \sum_{i \setminus \{j,k\}} |d_i^* - z_i| + 1 - z_j + z_k - d_k^* - 1 \quad (\textit{by the definition of S2}) \\
&= \sum_{i \setminus \{j,k\}} |d_i^* - z_i| + |z_j - d_j^*| + |z_k - d_k^*| \\
&= \|d^* - z\|
\end{aligned}
$$

Result 4 (Likelihood equation) *Given a connected graph G with degree sequence $\hat{d}$ (output of the Algorithm 2 is a graph with degree sequence $\hat{d}$), there exists a unique $\hat{\beta}(\hat{d})$ (MLE of β using $\hat{d}$) which satisfies the following equation*

$$\hat{d}_i = \sum_{j \neq i} \frac{e^{\beta_i + \beta_j}}{1 + e^{\beta_i + \beta_j}}$$

Proof 4 (Likelihood equation) *First of all we know that there is no affine transformations between the parameters β, and each edge is modeled with Bernoulli random variable with a specific probability function. Therefore, the likelihood function is from a minimal exponential family. Additionally, the parameters don't have non-linear relationships as well, so we have indeed full-rank minimal exponential family. Therefore, parameter space is convex and likelihood function is strictly concave. Additionally, $\hat{d}$ is from connected graph and not all degrees are same. Hence, likelihood equation provides unique MLE, $\hat{\beta}(\hat{d})$ under some extra conditions stated in Result 6 below. We need those extra conditions because the degree sequence must also satisfy convex hull properties. Now, we need to find the likelihood equation.*

We know that there is an edge (A_{ij}) between nodes i and j (independently) with

$$A_{ij} = 1 \quad w.p. \quad p_{ij} = \frac{e^{\beta_i + \beta_j}}{1 + e^{\beta_i + \beta_j}}, \quad i < j$$

Therefore,

$$\mathcal{L}(\beta_i) = \prod_{j\neq i} 1_{\{A_{ij}=1\}} p_{ij} + 1_{\{A_{ij}=0\}}(1-p_{ij}) \quad \textit{for all } i$$
$$= \prod_{j\neq i} \frac{e^{A_{ij}(\beta_i+\beta_j)}}{1+e^{\beta_i+\beta_j}} \quad \textit{for all } i$$

Then, the log-likelihood becomes

$$\log \mathcal{L}(\beta_i) = \sum_{j\neq i} A_{ij}(\beta_i+\beta_j) - \log(1+e^{\beta_i+\beta_j})$$

Now, take the derivative w.r.t β_i, and set it equal to 0.

$$\frac{\partial \log \mathcal{L}(\beta_i)}{\partial \beta_i} = \sum_{j\neq i} A_{ij} - \frac{e^{\beta_i+\beta_j}}{1+e^{\beta_i+\beta_j}} \overset{set}{=} 0$$
$$= \hat{d}_i - \sum_{j\neq i} \frac{e^{\beta_i+\beta_j}}{1+e^{\beta_i+\beta_j}} \overset{set}{=} 0$$
$$\rightarrow \hat{d}_i = \sum_{j\neq i} \frac{e^{\beta_i+\beta_j}}{1+e^{\beta_i+\beta_j}}$$

Following result is related to the consistency of estimator $\hat{\beta}(\hat{d})$. Proof of this result is similar to the proof of Lemma 4.2 at Chatterjee et al. [2011]. In the following proof, the parts proved at Chatterjee et al. [2011] is skipped to keep the discussion short.

Result 5 (Uniform consistency) *Let $\hat{\beta}(\hat{d})$ be a solution to the above likelihood equation, and $\epsilon = \Omega(\frac{1}{\sqrt{\log n}})$, then $\hat{\beta}(\hat{d})$ satisfies following inequality.*

$$\mathbb{P}(\max_i |\hat{\beta}_i(\hat{d}) - \beta_i| \leq C(L)\sqrt{\frac{\log n}{n}}) \geq 1 - C(L)n^{-2}$$

where $L = \max_i |\beta_i|$ and $C(L)$ is constant which only depends on L.

Proof 5 (Uniform consistency)
The proof is detailed and we need lemmas and propositions to be able to prove above result. First lemma shows that MLE of degree sequence (output from Algorithm 2 which is validated roughly) is asymptotically similar to the original degree sequence.

Lemma 1 (Similarity between $\hat{d}$ and d) *Let G be a random graph with degree sequence d, and let $\hat{d}$ is output from Algorithm 2, and z be the noisy version of d which is corrupted with discrete Laplacian noise. Then, for $\epsilon > \frac{1}{\sqrt{n}}$, and for sufficiently large n, the following equation is true.*

$$\mathbb{P}(\max_i |\hat{d}_i - d_i| > \sqrt{6n\log n}) \leq \frac{1}{n^2}$$

Proof 6 (Similarity between $\hat{d}$ and d) *As we have mentioned before, $\hat{d} \leq z$ due to the structure of the algorithm. This implies that $\max_i |\hat{d}_i - d_i| \leq \max_i |z_i - d_i|$. Also, define $|e_i| = |z_i - d_i|$ where e_i is an independent observation from discrete Laplace distribution. Lets say this this distribution has following p.m.f*

$$P(e_i = e) = \frac{1-p}{1+p} p^{|e|}, \quad 0 < p < 1, e \in \mathbb{Z}$$

First of all, the claim is that $\mathbb{P}(\max_i |e_i| \geq c) = 1 - (1 - \frac{2p^{[c]}}{1+p})^n$. This comes from the definition of p.m.f. In detail, firstly, we have

$$\begin{aligned}
\mathbb{P}(-c < e < c) &= \sum_{i=-[c]}^{[c]} \frac{1-p}{1+p} p^{|i|} \\
&= 1 - \sum_{i=-\infty}^{[c]} \frac{1-p}{1+p} p^{|i|} - \sum_{i=[c]}^{\infty} \frac{1-p}{1+p} p^{|i|} \\
&= 1 - \frac{1-p}{1+p} \frac{2p^{[c]}}{1-p} \\
&= 1 - \frac{2p^{[c]}}{1+p}
\end{aligned}$$

Then we have

$$\mathbb{P}(\max_i |e_i| \geq c) = 1 - \left(1 - \frac{2p^{[c]}}{1+p}\right)^n$$

Hence,

$$\begin{aligned}
\mathbb{P}(\max_i |\hat{d}_i - d_i| \geq c) &\leq \mathbb{P}(\max_i |z_i - d_i| \geq c) \\
&= \mathbb{P}(\max_i |e_i| \geq c) \\
&= 1 - 1 - \left(1 - \frac{2p^{[c]}}{1+p}\right)^n \\
&= 1 - \left(1 - \frac{2e^{-c\epsilon/2}}{1+e^{-\epsilon/2}}\right)^n \quad \textit{(since } p = e^{-\epsilon/2} \textit{ in the noise mechanism)} \\
&= 1 - (1 - k_n e^{-c\epsilon/2})^n, \quad \textit{where } k_n = \frac{2}{1+e^{-\epsilon/2}} \\
&= k_n e^{-c\epsilon/2} \textit{poly}(e^{-c\epsilon/2})
\end{aligned}$$

Last inequality comes from the fact that $a^n - b^n = (a-b)poly(a,b)$ where poly is some polynomial function. When $a = 1$ and $b = 1 - k_n e^{-c\epsilon/2}$, we obtain the above result.

Now, we have

$$\mathbb{P}(\max_i |\hat{d}_i - d_i| \geq c) \leq k_n e^{-c\epsilon/2} \textit{poly}(e^{-c\epsilon/2})$$

There are two cases. If poly$(e^{-c\epsilon/2}) \leq 0$, then $\mathbb{P}(\max_i |\hat{d}_i - d_i| \geq c) \leq 0 \leq \frac{1}{n^2}$. If poly$(e^{-c\epsilon/2}) > 0$, then by letting $c = [\sqrt{6n\log n}]$ and $n \to \infty$, we have

$$\mathbb{P}(\max_i |\hat{d}_i - d_i| \geq [\sqrt{6n\log n}]) \leq k_n poly(e^{-\sqrt{6\log n}/2}) \quad \textit{when } \epsilon > \frac{1}{\sqrt{n}}$$
$$\leq \frac{1}{n^2}$$

where the last inequality comes from the fact that poly$(e^{-\sqrt{6\log n}/2}) = o(1)$ and $k_n = O(1)$. The reason is that k_n is increasing function with respect to ϵ. When $\epsilon \to \infty$, $k_n \to 2$, so $k_n \leq 2$. Then, $O(1)o(1) = o(1)$. Also, $e^{-\sqrt{6\log n}/2}$ shrinks much faster than $\frac{1}{n^2}$ because it is also a polynomial term. Then, $k_n$poly$(e^{-\sqrt{6\log n}/2}) \leq \frac{1}{n^2}$ when n is large and $\epsilon > \frac{1}{\sqrt{n}}$.

Now, we provide some results from another paper. No proofs are provided for them in this paper as they can be found in Chatterjee et al. [2011].

Result 6 (from Chatterjee et al. [2011])
In Chatterjee et al. [2011], this has been proven with d and $\tilde{d}$ where $\tilde{d}$ is any d satisfying likelihood equation above. Instead of d, this time we replace it with $\hat{d}$. Let G be a random graph with degree sequence d, and $\hat{d}$ be the output from Algorithm 2. Let $L = \max_i |\beta_i|$ and $c \in (0,1)$. Then there exists constants $C > 0$ and $c_1, c_2 \in (0,1)$ depending only on L. There also exists c_3 depending on L and c. Then, with probability at least $1 - 3n^{-2}$, the following events hold.

1. $E1 = \{c_2(n-1) \leq \hat{d}_i \leq c_1(n-1) \quad \forall i\}$
2. $E2 = \{\frac{1}{n^2} g(\hat{d}, B) \geq c_3 - \sqrt{\frac{6\log n}{n}}\}$

where $g(d, B)$ is some function of degree sequence and a subset of indexes B. See Lemma 4.1. from Chatterjee et al. [2011] for more details.

Another results from the same paper are like following. The first one provides a condition for existence of solution for likelihood equation. Whereas the second one shows the uniquness and bound for that solution.

1. *Let d be a point inside convex hull of DS_n. Suppose there exists $c_1, c_2 \in (0,1)$ such that $c_2(n-1) \leq d_i \leq c_1(n-1)$ for all i. Suppose c_3 is a positive constant such that*
$$\{\frac{1}{n^2}\tilde{g}(d, B) \geq c_3 - \sqrt{\frac{6\log n}{n}}\}$$
where $\tilde{g}(d, B)$ is $g(d, B)$ constructed with c_2. Then a solution $\hat{\beta}$ to the maximum likelihood equations (stated above) exists and $|\hat{\beta}|_\infty \leq c_4$ for c_4 only dependent on c_1, c_2, c_3.

2. *Suppose solution exists, then $\hat{\beta}$ must be unique and for any $\beta \in \mathbb{R}^n$, we have*
$$|\beta - \hat{\beta}|_\infty \leq C|\beta - x|_\infty$$
where $x_i = \log d_i - \log \sum_{j\neq i} r_{ij}(x)$ for $r_{ij}(x) = \frac{1}{e^{-x_j} + e^{x_i}}$ iteratively.

Now, we can prove the consistency.

1. *$\hat{d}$ is output of Algorithm 2, so by design it is in the interior of convex hull of DS_n. Therefore, it satisfies the conditions above under the events $E1, E2$. Therefore, solutions for likelihood equation exists and $|\hat{\beta}|_\infty \leq C(L)$.*

2. *Since the solution for likelihood equation exists, let's say $\bar{d}$, it is unique, again by the result above.*

3. *Using the same result $|\beta - \hat{\beta}|_\infty \leq C|\beta - x|_\infty$. Now, note that since $\hat{d}$ lies in the convex hull*

$$\begin{aligned}
|\beta_i - x_i|_\infty &= |\beta_i - (\log \hat{d}_i - \log \sum_{j\neq i} \frac{1}{e^{-\beta_j} + e^{\beta_i}})|_\infty \\
&= |\log e^{\beta_i} - (\log \hat{d}_i - \log \sum_{j\neq i} \frac{1}{e^{-\beta_j} + e^{\beta_i}})|_\infty \\
&= |\log \sum_{j\neq i} \frac{e^{\beta_i}}{e^{-\beta_j} + e^{\beta_i}} - \log \hat{d}_i|_\infty \\
&= |\log \sum_{j\neq i} \frac{e^{\beta_i+\beta_j}}{1 + e^{\beta_i+\beta_j}} - \log \hat{d}_i|_\infty \\
&= |\log \bar{d}_i - \log \hat{d}_i|_\infty
\end{aligned}$$

and for $n > C$ this is bounded in probability by $C(L)\sqrt{\frac{\log n}{n}}$ by using Hoeffding's inequality for bounded random variables. Here, a note for the reader is that there is no clear explanation where this upper bound comes from. However, it must be closely related to Hoeffding's inequality and common bounds for concentration inequalities. To remove the $n > C$ constraint, one can increase $C(L)$ until $1 - C(L)n^{-2} < 0$ when $n \leq C$, so that n also becomes large. Note that $1 - C(L)n^{-2} < 0$ comes from Lemma 1 (similarity between $\hat{d}$ and d. In detail,

$$1 - C(L)n^{-2} < 0 \to C(L) > \frac{1}{n^2}$$

Therefore, we found probabilistic bound such that

$$\mathbb{P}(\max_i |\hat{\beta}_i(\hat{d}) - \beta_i| \leq C(L)\sqrt{\frac{\log n}{n}}) \geq 1 - C(L)n^{-2}$$

0.4 Future directions

There are couple of issues regarding the method proposed in this paper. As we have discussed above, adding noise on degree sequences is problematic, and so, authors considered a graph recovering algorithm to mitigate these problems. However, this raises a concern because one needs to go over all the steps of algorithm (which is based on Havel-Hakimi decomposition) to find an MLE of degree sequence, which might be inefficient. Additionally, resulting graph recovered from noisy sequence is usually disconnected. This is because of the fact that noisy degree sequence might be negative due to the random noise. Authors came up with an iterative method to adjust resulting disconnected graph. However, this adjusted sequence needs to be checked if it is a graphical sequence, or if it satisfies the stated conditions for MLE existence. Instead, one can use truncated noise mechanism and add positive noise [Croft et al., 2022], or independent but not iid noise so that resulting sequence stays in the range. Moreover, the paper doesn't have any discussion on the solution method of the likelihood equation. Usually, iterative gradient based methods are utilized to solve this problem. However, these methods are usually very inefficient when the network is sparse (which is a better way to model real-life networks). Hence, one can consider regularized versions of the likelihood equation. Especially, $L2$ regularization makes it easier to estimate sparse network models with less computational burden [Shao et al., 2024].

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